DEFAULTER ANALYSIS AND CLASSIFICATION FOR A MONEY LENDING COMPANY

PROPOSAL

BDM CAPSTONE PROJECT

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Declaration Statement

I am working on a Project Titled “DEFAULTER ANALYSIS AND CLASSIFICATION FOR A MONEY LENDING COMPANY”. I extend my appreciation to Home Credit Group and Kaggle for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project is genuine and precise to the utmost extent of my knowledge and capabilities. I affirm that all procedures employed for data analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project’s completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Harshil Pandey

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Signature

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Executive Summary

The project focuses on an international non-bank financial company called Home Credit Group. It provides financial solutions like loans and EMI based payments to both individual customers and businesses which makes it a B2C as well as a B2B business.

Since the company's main objective is to provide financial solutions to those who have no or insufficient credit histories, the major problem which the business faces is to identify potential defaulters who fail to repay the loan amount either partially or completely. On the other hand, the loan approval criteria cannot be made too strict as the company may reject loans of valuable non-defaulter customers.

The issues will be addressed by analyzing the data through different analytical procedures like data cleaning, identifying important attributes for defaulter classification and tuning a machine learning model on the data. For the customers that do have some credit history, efforts to aggregate it and include past data in the analysis will also be made. Tools like MS Excel, Python and its various libraries will extensively be used for data preprocessing, analysis and visualization.

The expected outcome helps the company to classify potential defaulters accurately based on relevant attributes and past data if available which assists in minimizing losses.

Organization Background

Home Credit Group was founded in 1997 in the Czech Republic. It is a financial company that aims to provide credit solutions to individual consumers and businesses that have insufficient credit history. PPF, an international investment group based in the Czech Republic is a major shareholder of the Home Credit Group.

The company expanded to Slovakia in 1999 and as of 2024, Home Credit Group has major presence in Asian countries like Kazakhstan, China, India, Philippines, and Indonesia. In India itself, the company has over 5000 employees and has served over 1.6 Cr customers.

Services offered by Home Credit include personal loans, credit cards, home loans, and EMI based payments on products like mobile phones and home appliances. Although customers can visit the nearby partner shops for application process, they can also download the Home Credit Mobile Application to apply for loans and pay installments. Thus, the process is completely online and needs minimal documentation. All relevant information for the customers is provided through the mobile application but further information can be accessed through the website: https://www.homecredit.net/.

Problem Statement

The problem statement can be listed as 2 objectives:

1. Accurate defaulter classification: The company records huge number of customer attributes. Out of all the attributes, identifying the most important ones that lead to reliable classification is a challenge. Moreover, it has to be kept in mind that too many loans are not rejected.

2. Utilizing past data: Utilizing past data is also a major obstacle as not every customer has past data corresponding to them. Finding innovative ways to include such data into the analysis will be the main task.

Background of the Problem

Employment and credit history are among the most important factors that lenders consider for loan approvals. Unfortunately, not everyone has a strong economic and employment background and this gives rise to difficulty in getting loans from reputed banks. Small businesses, especially operating in unorganized sector also face this issue of credit exclusion. Home Credit Group offers attractive solutions to deal with these issues in the form of instant loan approvals requiring minimal documentation.

But due to lack of credit history, the company faces a major problem of identifying potential defaulters who later on fail to repay the borrowed money either partially or completely. This not only leads to losses but also a feeling of distrust between the company and its customers. Furthermore, the loan approval criteria cannot be made too strict as it may lead to loan rejection of customers who could have repaid the borrowed amount. So, it is essential for the company to not reject too many loans along with accurately classifying probable defaulters. To achieve this, identifying the most important customer attributes is crucial.

Some customers do have reasonable credit history in the form of previous transactional data or some of them being previous customers of the company. Utilizing this complex data into determining the repayment capability of the customer is also a major challenge for the company.

Problem Solving Approach

Methods Used

The data has already been collected by the company and stored in the form of various comma separated files. A data driven approach will be used to solve the problems mentioned above which can be divided into the following main steps:

1. Exploratory Data Analysis: This step will involve understanding the nature and meaning of different recorded customer attributes by performing basic descriptive statistics over them. This will act as an initial filter and aid in dropping those attributes that have highly skewed distributions or provide no insight when related to the repayment capability of the customer.

2. Cleaning: Cleaning will involve data manipulations like normalization and imputation of null values. Features having high percentage of null values may also be dropped as they may not be useful. Normalization makes sure that different features have the same scale and it helps in visualization and comparing different features. Imputation of null values prevents loss of valuable data and it is known that machine learning algorithms perform well with clean data as input.

3. Feature Selection and Engineering: In this step the features that are not dropped will be further explored and new features may be created using the already given features that may aid in revealing significant trends and more accurate classification. Efforts to fit a simple machine learning model on the transformed data will also be made to unlock the full potential of data.

4. Aggregating Past Data: Past data in the form of previous transactions and applications will be aggregated carefully in order to include it in the analysis. Such data may give an intuition on the behavior of the customer and may reveal meaningful insights.

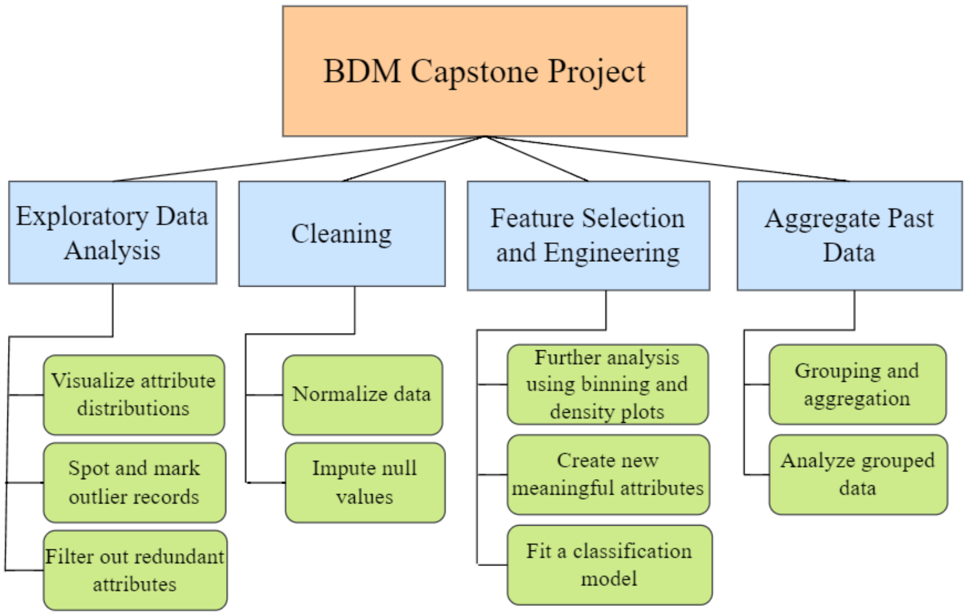
Analysis Tools

The principle analysis tool will be Python. Python is a general-purpose programming language that has tremendous support for data analysis tasks. For numerical computations and data manipulation, Python libraries like NumPy and Pandas will be used. These are powerful libraries that have useful functionalities to perform efficient analysis on large datasets. For visualization, libraries like Matplotlib and Seaborn will be used as they have a wide array of built-in plots and charts which provide full control over the visualization process. Scikit-Learn library may also be used which provides support to run various machine learning algorithms on structured data.

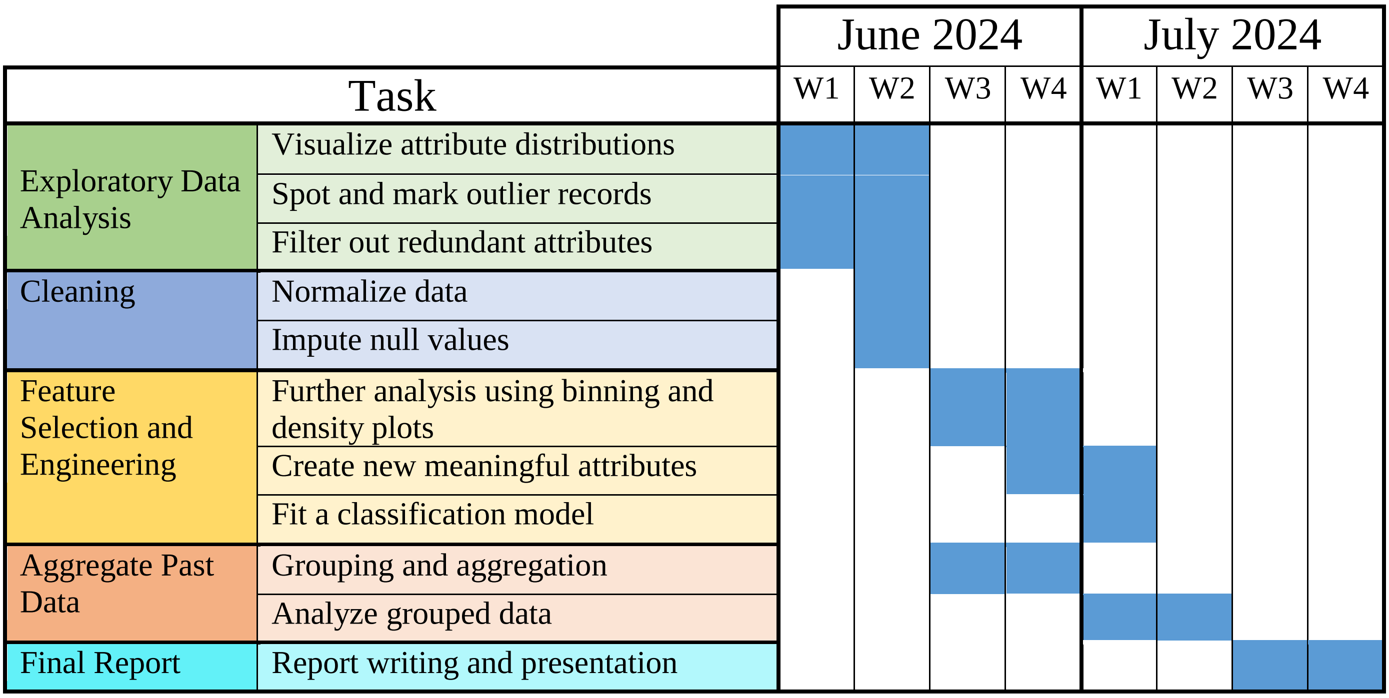
For analysis on smaller subsets of data, MS Excel may also be used since it is easy to work with and provides useful plotting and aggregation features like bar charts, histograms and pivot tables.

Expected Timeline

Work Breakdown Structure



Gantt Chart



Expected Outcomes

There are two expected outcomes of this project:

1. First expected outcome is that the company will be able to identify and classify potential defaulters more accurately, and also makes sure that it does not reject too many capable customers in the process.

2. Second expected outcome is that the company will be able to utilize the past transactional data of previous customers more meaningfully.

Both these outcomes may help the company to not only reduce its losses but also to achieve its main objective of credit inclusion.